

$$v(m, t), u(m, t), e(m, t) \quad (1)$$

$$\frac{\partial v}{\partial t} = B_0 \frac{\partial}{\partial m} (r^{n-1} u) \quad (2)$$

$$\frac{\partial u}{\partial t} = -A(r(m, t)) \frac{\partial p}{\partial m} \quad (3)$$

$$\frac{\partial e}{\partial t} = q - p \frac{\partial v}{\partial t} \quad (4)$$

NOTE! $\partial/\partial t$ here is a time derivative at constant m (which was shown to be identical to the material deriv. D/Dt)

In addition, the connection to the radial coordinate is via:

$$m(r, t) = \int^r \rho(r', t) A(r') dr' \quad (5)$$

$$\frac{\partial r}{\partial t} = u \quad (6)$$

The acceleration is defined by:

$$a(m, t) = -A(r(m, t)) \frac{\partial p}{\partial m} \quad (7)$$

so that:

$$\frac{\partial u}{\partial t} = a \quad (8)$$

1 Self Similar solution

planar:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (9)$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (10)$$

$$\frac{\partial e}{\partial t} + p \frac{\partial v}{\partial t} = 0 \quad (11)$$

$$pv = (\gamma - 1) e \quad (12)$$

$$p(0, t) = p_0 t^\tau \quad (13)$$

$$\rho(r, 0) = \rho_0 r^{-\omega} \quad (14)$$

$$v(r, 0) = v_0 r^\omega \quad (15)$$

$$v_0 = \frac{1}{\rho_0} \quad (16)$$

$$\left[\frac{E}{L} \right] = [p_0] [T^\tau] \quad (17)$$

$$\left[\frac{ML^2T^{-2}}{L} \right] = [p_0] [T^\tau] \quad (18)$$

$$[p_0] = MLT^{-\tau-2} \quad (19)$$

$$\left[\frac{L}{M} \right] = [v_0] [L^\omega] \quad (20)$$

$$[v_0] = [M^{-1}L^{1-\omega}] \quad (21)$$

p	u	v	v_0	p_0	m	t
MLT^{-2}	LT^{-1}	$M^{-1}L$	$M^{-1}L^{1-\omega}$	$MLT^{-\tau-2}$	M	T

we have $N = 7$ quantities in the problem, with $K = 3$ units. Hence we have $N - K = 4$ dimensionless quantities.

$$\xi = mv_0^a p_0^b t^c \quad (22)$$

no M units:

$$1 - a + b = 0 \quad (23)$$

no L units:

$$(1 - \omega) a + b = 0 \quad (24)$$

putting the first in the second:

$$(1 - \omega) (1 + b) + b = 0 \quad (25)$$

$$(2 - \omega) b = - (1 - \omega) \quad (26)$$

$$b = \frac{\omega - 1}{2 - \omega} \quad (27)$$

$$a = b + 1 = 1 + \frac{\omega - 1}{2 - \omega} \quad (28)$$

$$a = \frac{2 - \omega - 1 + \omega}{2 - \omega} \quad (29)$$

$$a = \frac{1}{2 - \omega} \quad (30)$$

no T units:

$$-(\tau + 2) b + c = 0 \quad (31)$$

$$c = \frac{(\tau + 2) (\omega - 1)}{2 - \omega} \quad (32)$$

hence:

$$\xi = mv_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (33)$$

$$V = vv_0^a p_0^b t^c \quad (34)$$

no M units:

$$-1 - a + b = 0 \quad (35)$$

no L units:

$$1 + (1 - \omega) a + b = 0 \quad (36)$$

putting the first in the second:

$$1 + (1 - \omega) a + 1 + a = 0 \quad (37)$$

$$(2 - \omega) a = -2 \quad (38)$$

$$a = \frac{2}{\omega - 2} \quad (39)$$

$$b = a + 1 = \frac{\omega}{\omega - 2} \quad (40)$$

not T units:

$$-(\tau + 2) b + c = 0 \quad (41)$$

$$c = (\tau + 2) b \quad (42)$$

$$c = \frac{(\tau + 2) \omega}{\omega - 2} \quad (43)$$

$$V = vv_0^{\frac{2}{\omega-2}} p_0^{\frac{\omega}{\omega-2}} t^{\frac{(\tau+2)\omega}{\omega-2}} \quad (44)$$

$$v(m, t) = V(\xi) v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (45)$$

$$U = uv_0^a p_0^b t^c \quad (46)$$

no M units:

$$-a + b = 0 \quad (47)$$

no L units:

$$1 + (1 - \omega) a + b = 0 \quad (48)$$

putting the first in the second:

$$1 + (1 - \omega) a + a = 0 \quad (49)$$

$$a = b = \frac{1}{\omega - 2} \quad (50)$$

not T units:

$$-1 - (\tau + 2) b + c = 0 \quad (51)$$

$$-1 - (\tau + 2) \frac{1}{\omega - 2} + c = 0 \quad (52)$$

$$c = 1 + (\tau + 2) \frac{1}{\omega - 2} \quad (53)$$

$$c = \frac{\omega + \tau}{\omega - 2} \quad (54)$$

hence:

$$U = uv_0^{\frac{1}{\omega-2}} p_0^{\frac{1}{\omega-2}} t^{\frac{\omega+\tau}{\omega-2}} \quad (55)$$

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (56)$$

$$P = pv_0^a p_0^b t^c \quad (57)$$

no M units:

$$1 - a + b = 0 \quad (58)$$

no L units:

$$1 + (1 - \omega) a + b = 0 \quad (59)$$

putting the first in the second:

$$1 + (1 - \omega) (b + 1) + b = 0 \quad (60)$$

$$(2 - \omega) (b + 1) = 0 \quad (61)$$

$$b = -1 \quad (62)$$

$$a = 0 \quad (63)$$

not T units:

$$-2 - (\tau + 2) b + c = 0 \quad (64)$$

$$-2 + (\tau + 2) + c = 0 \quad (65)$$

$$c = -\tau \quad (66)$$

$$P = \frac{p}{p_0 t^\tau} \quad (67)$$

$$p(m, t) = P(\xi) p_0 t^\tau \quad (68)$$

summary:

$$\xi = m v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (69)$$

$$v(m, t) = V(\xi) v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (70)$$

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (71)$$

$$p(m, t) = P(\xi) p_0 t^\tau \quad (72)$$

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (73)$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (74)$$

$$\frac{\partial}{\partial t} \left(\frac{pv}{r} \right) + p \frac{\partial v}{\partial t} = 0 \quad (75)$$

$$r = \gamma - 1 \quad (76)$$

1.1 Derivation of self similar ODE

$$\xi = m v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (77)$$

$$\frac{\partial \xi}{\partial m} = \frac{\xi}{m} = v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (78)$$

$$\frac{\partial}{\partial m} \left(\frac{\xi}{m} \right) = 0 \quad (79)$$

$$\frac{\partial \xi}{\partial t} = \frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{t} \quad (80)$$

The continuity PDE is:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (81)$$

now:

$$\frac{\partial v}{\partial t} = \frac{\partial}{\partial t} \left(V v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \right) = v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \left(V' \frac{\partial \xi}{\partial t} + \frac{V}{t} \frac{(\tau+2)\omega}{2-\omega} \right) \quad (82)$$

$$= v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \left(V' \frac{(\tau+2)(\omega-1)\xi}{2-\omega} + \frac{V}{t} \frac{(\tau+2)\omega}{2-\omega} \right) \quad (83)$$

$$= v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}-1} \left(\xi V' \frac{(\tau+2)(\omega-1)}{2-\omega} + V \frac{(\tau+2)\omega}{2-\omega} \right) \quad (84)$$

$$= v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+2\omega-2+\omega}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \quad (85)$$

$$= v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \quad (86)$$

now:

$$\frac{\partial u}{\partial m} = \frac{\partial}{\partial m} \left(U v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \right) \quad (87)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \frac{\partial \xi}{\partial m} U' \quad (88)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \frac{\partial \xi}{\partial m} U' \quad (89)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} U' \quad (90)$$

$$= v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\omega+\tau+\omega-\tau+2\omega-2}{2-\omega}} U' \quad (91)$$

$$= v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} U' \quad (92)$$

Putting it back in the PDE:

$$\frac{\partial}{\partial t} \left(V v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \right) - \frac{\partial}{\partial m} \left(U v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \right) = 0 \quad (93)$$

$$v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) - v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} U' = 0 \quad (94)$$

$$\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) - U' = 0 \quad (95)$$

$$\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) - U' = 0 \quad (96)$$

momentum equation:

$$\frac{\partial u}{\partial t} = -\frac{\partial p}{\partial m} \quad (97)$$

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t} \left(U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \right) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \left(U' \frac{\partial \xi}{\partial t} + \frac{U}{t} \left(\frac{\omega+\tau}{2-\omega} \right) \right) \quad (98)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \left(U' \frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{t} + \frac{U}{t} \left(\frac{\omega+\tau}{2-\omega} \right) \right) \quad (99)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}-1} \left(\xi U' \frac{(\tau+2)(\omega-1)}{2-\omega} + U \left(\frac{\omega+\tau}{2-\omega} \right) \right) \quad (100)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2\omega+\tau-2}{2-\omega}} \left(\xi U' \frac{(\tau+2)(\omega-1)}{2-\omega} + U \left(\frac{\omega+\tau}{2-\omega} \right) \right) \quad (101)$$

$$\frac{\partial p}{\partial m} = \frac{\partial}{\partial m} (P p_0 t^\tau) = p_0 t^\tau \frac{\partial P}{\partial m} = p_0 t^\tau P' \frac{\partial \xi}{\partial m} \quad (102)$$

$$= p_0 t^\tau P' v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega-1}{2-\omega} \frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (103)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}+1} t^{\frac{(\tau+2)(\omega-1)}{2-\omega} + \tau} P' \quad (104)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2\omega+\tau-2}{2-\omega}} P' \quad (105)$$

Putting it back in the PDE:

$$\frac{\partial u}{\partial t} = -\frac{\partial p}{\partial m} \quad (106)$$

$$v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2\omega+\tau-2}{2-\omega}} \left(\xi U' \frac{(\tau+2)(\omega-1)}{2-\omega} + U \left(\frac{\omega+\tau}{2-\omega} \right) \right) = -v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2\omega+\tau-2}{2-\omega}} P' \quad (107)$$

hence

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \xi U' + U \left(\frac{\omega+\tau}{2-\omega} \right) + P' = 0 \quad (108)$$

energy equation:

$$\frac{\partial}{\partial t} \left(\frac{pv}{r} \right) + p \frac{\partial v}{\partial t} = 0 \quad (109)$$

$$\frac{v}{r} \frac{\partial p}{\partial t} + \frac{p}{r} \frac{\partial v}{\partial t} + p \frac{\partial v}{\partial t} = 0 \quad (110)$$

$$\left(1 + \frac{1}{r} \right) p \frac{\partial v}{\partial t} = -\frac{v}{r} \frac{\partial p}{\partial t} \quad (111)$$

we already calculated $\frac{\partial v}{\partial t}$:

$$\frac{\partial v}{\partial t} = v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \quad (112)$$

now:

$$\frac{\partial p}{\partial t} = \frac{\partial}{\partial t} (P p_0 t^\tau) = p_0 t^\tau \left(P' \frac{\partial \xi}{\partial t} + \tau P \frac{1}{t} \right) \quad (113)$$

$$= p_0 t^\tau \left(P' \frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{t} + \tau P \frac{1}{t} \right) \quad (114)$$

$$= p_0 t^{\tau-1} \left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) \quad (115)$$

putting this in the energy equation:

$$\left(1 + \frac{1}{r} \right) p \frac{\partial v}{\partial t} = - \frac{v}{r} \frac{\partial p}{\partial t} \quad (116)$$

$$\left(1 + \frac{1}{r} \right) [P p_0 t^\tau] \left[v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \right] \quad (117)$$

$$= - \frac{1}{r} \left[V v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \right] \left[p_0 t^{\tau-1} \left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) \right] \quad (118)$$

$$\left(1 + \frac{1}{r} \right) [P] \left[t^{\frac{\tau\omega+3\omega-2}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \right] \quad (119)$$

$$= - \frac{1}{r} \left[V t^{\frac{(\tau+2)\omega}{2-\omega}} \right] \left[t^{-1} \left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) \right] \quad (120)$$

the temporal power on the right is:

$$\frac{(\tau+2)\omega}{2-\omega} - 1 = \frac{\tau\omega+2\omega-2+\omega}{2-\omega} = \frac{\tau\omega+3\omega-2}{2-\omega} \quad (121)$$

which is the same as in the lhs. Hence:

$$\left(1 + \frac{1}{r} \right) P \left[\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \right] \quad (122)$$

$$= - \frac{1}{r} V \left[\left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) \right] \quad (123)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{r} V P' + \frac{\tau}{r} P V + \left(1 + \frac{1}{r} \right) P \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) = 0 \quad (124)$$

1.2 summary

$$\xi = m v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (125)$$

$$v(m, t) = V(\xi) v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (126)$$

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (127)$$

$$p(m, t) = P(\xi) p_0 t^\tau \quad (128)$$

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (129)$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (130)$$

$$\frac{\partial}{\partial t} \left(\frac{pv}{r} \right) + p \frac{\partial v}{\partial t} = 0 \quad (131)$$

$$r = \gamma - 1 \quad (132)$$

the self similar hydro equations are:

$$\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) - U' = 0 \quad (133)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \xi U' + U \left(\frac{\omega+\tau}{2-\omega} \right) + P' = 0 \quad (134)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{r} V P' + \frac{\tau}{r} P V + \left(1 + \frac{1}{r} \right) P \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) = 0 \quad (135)$$

2 canonical form of the ODE

now we want to diagonalize and get a canonical form of equations $\mathbf{y}' = \mathbf{f}(\mathbf{y})$. It is possible since this is a linear equation in terms of V', U', P' .

$$\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) - U' = 0 \quad (136)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \xi U' + U \left(\frac{\omega+\tau}{2-\omega} \right) + P' = 0 \quad (137)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{r} VP' + \frac{\tau}{r} PV + \left(1 + \frac{1}{r}\right) P \left(\frac{\tau+2}{2-\omega}\right) ((\omega-1)\xi V' + \omega V) = 0 \quad (138)$$

write it as $Ax = b$

$$\frac{(\tau+2)(1-\omega)}{2-\omega} \frac{\xi}{r} \xi V' + U' = V \frac{\omega(\tau+2)}{2-\omega} \quad (139)$$

$$\frac{(\tau+2)(1-\omega)}{2-\omega} \xi U' - P' = U \left(\frac{\omega+\tau}{2-\omega}\right) \quad (140)$$

third equation:

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{r} VP' + \frac{\tau}{r} PV + \left(1 + \frac{1}{r}\right) P \left(\frac{\tau+2}{2-\omega}\right) ((\omega-1)\xi V' + \omega V) = 0 \quad (141)$$

$$\left(1 + \frac{1}{r}\right) P \frac{(\tau+2)(1-\omega)}{2-\omega} \xi V' + \frac{(\tau+2)(1-\omega)}{2-\omega} \frac{\xi}{r} VP' = PV \left(\frac{\tau}{r} + \left(1 + \frac{1}{r}\right) \frac{(\tau+2)\omega}{2-\omega}\right) \quad (142)$$

multiply by r :

$$(1+r) \frac{(\tau+2)(1-\omega)}{2-\omega} \xi PV' + \frac{(\tau+2)(1-\omega)}{2-\omega} \xi VP' = PV \left(\tau + (1+r) \frac{(\tau+2)\omega}{2-\omega}\right) \quad (143)$$

hence:

$$\left(\frac{\tau+2}{2-\omega}\right) ((\omega-1)\xi V' + \omega V) - U' = 0 \quad (144)$$

$$\frac{(\tau+2)(1-\omega)}{2-\omega} \xi U' - P' = U \left(\frac{\omega+\tau}{2-\omega}\right) \quad (145)$$

$$(1+r) \frac{(\tau+2)(1-\omega)}{2-\omega} \xi PV' + \frac{(\tau+2)(1-\omega)}{2-\omega} \xi VP' = PV \left(\tau + (1+r) \frac{(\tau+2)\omega}{2-\omega}\right) \quad (146)$$

define

$$a = \frac{(\tau+2)(1-\omega)}{2-\omega} \xi \quad (147)$$

$$c_1 = V \frac{\omega(\tau+2)}{2-\omega} \quad (148)$$

$$c_2 = U \left(\frac{\omega+\tau}{2-\omega}\right) \quad (149)$$

$$c_3 = PV \left(\tau + (1+r) \frac{(\tau+2)\omega}{2-\omega}\right) \quad (150)$$

so we get:

$$aV' + U' = c_1 \quad (151)$$

$$aU' - P' = c_2 \quad (152)$$

$$(1 + r) aPV' + aVP' = c_3 \quad (153)$$

define:

$$b = (1 + r) aP \quad (154)$$

$$d = aV \quad (155)$$

we get:

$$aV' + U' = c_1 \quad (156)$$

$$aU' - P' = c_2 \quad (157)$$

$$bV' + dP' = c_3 \quad (158)$$

according to wolfram alpha the solution is:

$$V' = \frac{\Delta_1}{\Delta}, \quad U' = \frac{\Delta_2}{\Delta}, \quad P' = \frac{\Delta_3}{\Delta} \quad (159)$$

$$\Delta = a^2d - b \quad (160)$$

$$\Delta_1 = c_1ad - c_2d - c_3 \quad (161)$$

$$\Delta_2 = -c_1b + c_2ad + c_3a \quad (162)$$

$$\Delta_3 = -c_1ab + c_2b + c_3a^2 \quad (163)$$

when

$$\Delta \neq 0 \quad (164)$$

2.1 using $\rho = 1/v$

when $\omega < 0$, we may have $v \rightarrow \infty$ for $m \rightarrow 0$. So we may want to work with $\rho = 1/v$ instead.

In this case, define

$$R(\xi) = \frac{1}{V(\xi)} \quad (165)$$

$$V' = \frac{1}{R} \quad (166)$$

$$V' = -\frac{1}{R^2} R' \quad (167)$$

so if we had:

$$V' = \frac{\Delta_1(V, U, P)}{\Delta(V, U, P)}, \quad U' = \frac{\Delta_2(V, U, P)}{\Delta(V, U, P)}, \quad P' = \frac{\Delta_3(V, U, P)}{\Delta(V, U, P)} \quad (168)$$

so we have

$$R' = -R^2 \frac{\Delta_1\left(\frac{1}{R}, U, P\right)}{\Delta\left(\frac{1}{R}, U, P\right)}, \quad U' = \frac{\Delta_2\left(\frac{1}{R}, U, P\right)}{\Delta\left(\frac{1}{R}, U, P\right)}, \quad P' = \frac{\Delta_3\left(\frac{1}{R}, U, P\right)}{\Delta\left(\frac{1}{R}, U, P\right)} \quad (169)$$

we had:

$$\Delta = a^2 d - b \quad (170)$$

$$\Delta_1 = c_1 a d - c_2 d - c_3 \quad (171)$$

$$\Delta_2 = -c_1 b + c_2 a d + c_3 a \quad (172)$$

$$\Delta_3 = -c_1 a b + c_2 b + c_3 a^2 \quad (173)$$

with

$$a = \frac{(\tau + 2)(1 - \omega)}{2 - \omega} \xi, \quad b = (r + 1) a P, \quad d = a V \quad (174)$$

$$c_1 = V \frac{\omega(\tau + 2)}{2 - \omega} \quad (175)$$

$$c_2 = U \left(\frac{\omega + \tau}{2 - \omega} \right) \quad (176)$$

$$c_3 = P V \left(\tau + (1 + r) \frac{(\tau + 2)\omega}{2 - \omega} \right) \quad (177)$$

We want to put the V factor explicitly in the rhs. Rewriting c_1, c_3, d without the V factor:

$$a = \frac{(\tau + 2)(1 - \omega)}{2 - \omega} \xi, \quad b = (1 + r) a P, \quad d = a \quad (178)$$

$$c_1 = \frac{\omega(\tau + 2)}{2 - \omega} \quad (179)$$

$$c_2 = U \left(\frac{\omega + \tau}{2 - \omega} \right) \quad (180)$$

$$c_3 = P \left(\tau + (1 + r) \frac{(\tau + 2)\omega}{2 - \omega} \right) \quad (181)$$

we get:

$$\Delta = a^3 V - b \quad (182)$$

$$\Delta_1 = c_1 a^2 V^2 - c_2 a V - c_3 V \quad (183)$$

$$\Delta_2 = -c_1 bV + c_2 a^2 V + c_3 aV \quad (184)$$

$$\Delta_3 = -c_1 abV + c_2 b + c_3 a^2 V \quad (185)$$

$$R' = -R^2 \frac{\Delta_1 \left(\frac{1}{R}, U, P \right)}{\Delta \left(\frac{1}{R}, U, P \right)} \quad (186)$$

$$R' = -R^2 \frac{c_1 a^2 V^2 - c_2 aV - c_3 V}{a^3 V - b} \quad (187)$$

$$R' = -R^2 \frac{c_1 ad \left(\frac{1}{R} \right)^2 - c_2 d \frac{1}{R} - c_3 \frac{1}{R}}{a^3 \frac{1}{R} - b} \quad (188)$$

$$R' = -\frac{c_1 ad - c_2 dR - c_3 R}{a^3 \frac{1}{R} - b} \quad (189)$$

$$R' = -\frac{R (c_1 a^2 - c_2 dR - c_3 R)}{a^3 - bR} \quad (190)$$

$$U' = \frac{\Delta_2}{\Delta} = \frac{-c_1 bV + c_2 a^2 V + c_3 aV}{a^3 V - b} \quad (191)$$

$$= \frac{\frac{1}{R} (-c_1 b + c_2 a^2 + c_3 a)}{a^3 \frac{1}{R} - b} \quad (192)$$

$$= \frac{-c_1 b + c_2 a^2 + c_3 a}{a^3 - bR} \quad (193)$$

$$P' = \frac{\Delta_3}{\Delta} = \frac{-c_1 abV + c_2 b + c_3 a^2 V}{a^3 V - b} \quad (194)$$

$$= \frac{-c_1 ab \frac{1}{R} + c_2 b + c_3 a^2 \frac{1}{R}}{a^3 \frac{1}{R} - b} \quad (195)$$

$$= \frac{-c_1 ab + c_2 bR + c_3 a^2}{a^3 - bR} \quad (196)$$

summary:

$$a = \frac{(\tau+2)(1-\omega)}{2-\omega}\xi, \quad b = (1+r)aP \quad (197)$$

$$c_1 = \frac{\omega(\tau+2)}{2-\omega}, \quad c_2 = U\left(\frac{\omega+\tau}{2-\omega}\right), \quad c_3 = P\left(\tau + (1+r)\frac{(\tau+2)\omega}{2-\omega}\right) \quad (198)$$

$$R' = \frac{-R(c_1a^2 - c_2aR - c_3R)}{a^3 - bR} \quad (199)$$

$$U' = \frac{-c_1b + c_2a^2 + c_3a}{a^3 - bR} \quad (200)$$

$$P' = \frac{-c_1ab + c_2bR + c_3a^2}{a^3 - bR} \quad (201)$$

or:

$$V' = \frac{V(c_1a^2V - c_2a - c_3)}{a^3V - b} \quad (202)$$

$$U' = \frac{V(-c_1b + c_2a^2 + c_3a)}{a^3V - b} \quad (203)$$

$$P' = \frac{-c_1abV + c_2b + c_3a^2V}{a^3V - b} \quad (204)$$

THE PROBLEM HERE IS THAT $a, b \propto \xi$ both goes to zero at $\xi \rightarrow 0$.

Now we want to have the ξ dependence

Now we take a, b without the ξ factor:

$$a = \frac{(\tau+2)(1-\omega)}{2-\omega}, \quad b = (1+r)aP \quad (205)$$

define

$$X = \frac{\xi}{R} = \xi V \rightarrow 0 \quad (206)$$

which goes to 0 at $\xi \rightarrow 0$.

$$R' = \frac{-R(c_1a^2\xi^2 - c_2a\xi R - c_3R)}{a^3\xi^3 - b\xi R} \quad (207)$$

$$= \frac{-\left(c_1a^2\xi - c_2aR - c_3\frac{R}{\xi}\right)}{a^3\xi\frac{\xi}{R} - b} \quad (208)$$

$$= \frac{-(c_1a^2\xi - c_2aR - \frac{c_3}{X})}{a^3\xi X - b} \quad (209)$$

Assuming $R \rightarrow 0$ as $\xi \rightarrow 0$, it seems like $R' \rightarrow -\infty$ for $\xi \rightarrow 0$:(

for $\xi \rightarrow 0$, we can write this eq. as:

$$R' \approx -\frac{c_3}{b} \frac{R}{\xi} \quad (210)$$

which has a power law solution

$$R(\xi) = A\xi^{-\frac{c_3}{b}} \quad (211)$$

$$\frac{c_3}{b} = \frac{\tau + (r+1) \frac{(\tau+2)\omega}{2-\omega}}{(r+1) \frac{(\tau+2)(1-\omega)}{2-\omega}} = \frac{\tau(2-\omega)}{(r+1)(\tau+2)(1-\omega)} + \frac{\omega}{1-\omega} \quad (212)$$

It seems like the numerical soltuion $R(\xi)$ agrees well with this power law near $\xi \rightarrow 0$.

Now

$$U' = \frac{-c_1 b \xi + c_2 a^2 \xi^2 + c_3 a \xi}{a^3 \xi^3 - b \xi R} \quad (213)$$

$$= \frac{-c_1 b + c_2 a^2 \xi + c_3 a}{a^3 \xi^2 - b R} \quad (214)$$

$$= \frac{-c_1 b \frac{1}{R} + c_2 a^2 \frac{\xi}{R} + c_3 a \frac{1}{R}}{a^3 \xi \frac{\xi}{R} - b} \quad (215)$$

$$= \frac{-c_1 b \frac{1}{R} + c_2 a^2 X + c_3 a \frac{1}{R}}{a^3 \xi X - b} \quad (216)$$

it seems like $U' \rightarrow -\infty$ for $\xi \rightarrow 0$:(

$$P' = \frac{-c_1 a b \xi^2 + c_2 b \xi R + c_3 a^2 \xi^2}{a^3 \xi^3 - b \xi R} \quad (217)$$

$$= \frac{-c_1 a b \xi + c_2 b R + c_3 a^2 \xi}{a^3 \xi^2 - b R} \quad (218)$$

$$= \frac{-c_1 a b \frac{\xi}{R} + c_2 b + c_3 a^2 \frac{\xi}{R}}{a^3 \xi \frac{\xi}{R} - b} \quad (219)$$

$$= \frac{-c_1 a b X + c_2 b + c_3 a^2 X}{a^3 \xi X - b} \quad (220)$$

which is finite as $\xi \rightarrow 0$:)

Now in terms of V :

$$V' = \frac{V(c_1 a^2 V \xi^2 - c_2 a \xi - c_3)}{a^3 V \xi^3 - b \xi} \quad (221)$$

$$= \frac{V(c_1 a^2 V \xi^2 - c_2 a \xi - c_3)}{\xi(a^3 V \xi^2 - b)} \quad (222)$$

$$= \frac{\frac{V}{\xi}(c_1 a^2 V \xi^2 - c_2 a \xi - c_3)}{a^3 \xi X - b} \quad (223)$$

$$= \frac{c_1 a^2 V^2 \xi - c_2 a V - \frac{V}{\xi} c_3}{a^3 \xi X - b} \quad (224)$$

$$= \frac{c_1 a^2 V X - c_2 a V - \frac{V}{\xi} c_3}{a^3 \xi X - b} \quad (225)$$

$$U' = \frac{V(-c_1 b \xi + c_2 a^2 \xi^2 + c_3 a \xi)}{a^3 V \xi^3 - b \xi} \quad (226)$$

$$= \frac{V(-c_1 b + c_2 a^2 \xi + c_3 a)}{a^3 \xi X - b} \quad (227)$$

$$= \frac{-c_1 b V + c_2 a^2 X + c_3 a V}{a^3 \xi X - b} \quad (228)$$

$$P' = \frac{-c_1 a b V \xi^2 + c_2 b \xi + c_3 a^2 V \xi^2}{a^3 V \xi^3 - b \xi} \quad (229)$$

$$= \frac{\frac{1}{\xi}(-c_1 a b V \xi^2 + c_2 b \xi + c_3 a^2 V \xi^2)}{a^3 \xi X - b} \quad (230)$$

$$= \frac{-c_1 a b X + c_2 b + c_3 a^2 X}{a^3 \xi X - b} \quad (231)$$

summary

Canonical equations in terms of ξ, R or ξ, V :

$$a = \frac{(\tau + 2)(1 - \omega)}{2 - \omega}, \quad b = (1 + r) a P \quad (232)$$

$$c_1 = \frac{\omega(\tau + 2)}{2 - \omega}, \quad c_2 = U \left(\frac{\omega + \tau}{2 - \omega} \right), \quad c_3 = P \left(\tau + (1 + r) \frac{(\tau + 2)\omega}{2 - \omega} \right) \quad (233)$$

$$R' = \frac{-(c_1 a^2 \xi - c_2 a R - \frac{c_3}{X})}{a^3 \xi X - b} \quad (234)$$

$$U' = \frac{-c_1 b \frac{1}{R} + c_2 a^2 X + c_3 a \frac{1}{R}}{a^3 \xi X - b} \quad (235)$$

$$P' = \frac{-c_1 a b X + c_2 b + c_3 a^2 X}{a^3 \xi X - b} \quad (236)$$

or:

$$V' = \frac{V \left(c_1 a^2 X - c_2 a - \frac{c_3}{\xi} \right)}{a^3 \xi X - b} \quad (237)$$

$$U' = \frac{-c_1 b V + c_2 a^2 X + c_3 a V}{a^3 \xi X - b} \quad (238)$$

$$P' = \frac{-c_1 a b X + c_2 b + c_3 a^2 X}{a^3 \xi X - b} \quad (239)$$

2.2 working with $\ln \xi$

$$\frac{df}{d \ln x} = \frac{df}{dx} \frac{dx}{d \ln x} = x \frac{df}{dx} \quad (240)$$

$$\frac{d \ln f}{d \ln x} = x \frac{d \ln f}{dx} = \frac{x}{f} \frac{df}{dx} \quad (241)$$

hence:

$$\frac{df}{dx} = \frac{1}{x} \frac{df}{d \ln x} \quad (242)$$

$$\frac{df}{dx} = \frac{f}{x} \frac{d \ln f}{d \ln x} \quad (243)$$

3 initial density $v(m, t = 0)$

$$m(r, t = 0) = \int_0^r \rho(r', t = 0) dr' \quad (244)$$

$$m(r, t = 0) = \int_0^r \rho_0 r'^{-\omega} dr' \quad (245)$$

$$m(r, t = 0) = \rho_0 \left[\frac{r^{1-\omega}}{1-\omega} \right]_0^r \quad (246)$$

we must have

$$\omega < 1 \quad (247)$$

$$m(r, t = 0) = \rho_0 \frac{r^{1-\omega}}{1-\omega} \quad (248)$$

$$\frac{m(1-\omega)}{\rho_0} = r^{1-\omega} \quad (249)$$

$$v_0 m(1-\omega) = r^{1-\omega} \quad (250)$$

$$r = [v_0(1-\omega)m]^{\frac{1}{1-\omega}} \quad (251)$$

which has units of length as check.

hence:

$$v(r, 0) = v_0 r^\omega = v_0 [v_0(1-\omega)m]^{\frac{\omega}{1-\omega}} \quad (252)$$

$$v(m, 0) = v_0^{\frac{\omega}{1-\omega}+1} [(1-\omega)m]^{\frac{\omega}{1-\omega}} \quad (253)$$

$$v(m, 0) = v_0^{\frac{1}{1-\omega}} [(1-\omega)m]^{\frac{\omega}{1-\omega}} \quad (254)$$

which, as check has units of $M^{-1}L$, since:

$$[v_0] = M^{-1}L^{1-\omega} \quad (255)$$

$$-\frac{1}{1-\omega} + \frac{\omega}{1-\omega} = -1 \quad (256)$$

4 Jump conditions Eulerian??

$$u_0 = p_0 = 0 \quad (257)$$

continuity:

$$D(\rho_1 - \rho_0) = u_1 \rho_1 - u_0 \rho_0 \quad (258)$$

$$D(\rho_1 - \rho_0) = u_1 \rho_1 \quad (259)$$

$$D\left(\frac{1}{v_1} - \frac{1}{v_0}\right) = \frac{u_1}{v_1} \quad (260)$$

$$\frac{D - u_1}{v_1} = \frac{D}{v_0} \quad (261)$$

momentum:

$$D(\rho_1 u_1 - \rho_0 u_0) = \rho_1 u_1^2 + p_1 - \rho_0 u_0^2 - p_0 \quad (262)$$

$$D\rho_1 u_1 = \rho_1 u_1^2 + p_1 \quad (263)$$

$$\frac{Du_1}{v_1} = \frac{u_1^2}{v_1} + p_1 \quad (264)$$

$$\frac{u_1(D - u_1)}{v_1} = p_1 \quad (265)$$

energy:

$$D\left(\left(e_1 + \frac{1}{2}u_1^2\right) - \left(e_0 + \frac{1}{2}u_0^2\right)\right) = u_1\left(\rho_1\left(e_1 + \frac{1}{2}u_1^2\right) + p_1\right) - u_0\left(\rho_0\left(e_0 + \frac{1}{2}u_0^2\right) + p_0\right) \quad (266)$$

$$D\rho_1\left(e_1 + \frac{1}{2}u_1^2\right) = u_1\left(\rho_1\left(e_1 + \frac{1}{2}u_1^2\right) + p_1\right) \quad (267)$$

$$p_1 u_1 = D\rho_1\left(e_1 + \frac{1}{2}u_1^2\right) - u_1\rho_1\left(e_1 + \frac{1}{2}u_1^2\right) \quad (268)$$

$$p_1 u_1 = \rho_1\left(e_1 + \frac{1}{2}u_1^2\right)(D - u_1) \quad (269)$$

$$p_1 u_1 v_1 = \left(e_1 + \frac{1}{2}u_1^2\right)(D - u_1) \quad (270)$$

5 Jump conditions Lagrangian

unshocked region:

$$v_{ref}, u_{ref} = 0, p_{ref} = 0 \quad (271)$$

shocked region:

$$v_s, u_s, p_s \quad (272)$$

reminder, for any conservation law of the form $\mathbf{u}_t + \mathbf{F}(\mathbf{u})_y = 0$, the Hugoniot jump conditions are $\dot{y}\Delta u = \Delta F$, where $\dot{y}$ is the speed of discontinuity.

continuity equation

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (273)$$

Hugoniot:

$$\dot{m}(v_s - v_{ref}) = u_{ref} - u_s \quad (274)$$

$$\frac{D}{v_0}(v_s - v_{ref}) = -u_s \quad (275)$$

$$D(v_s - v_{ref}) = -v_{ref}u_s \quad (276)$$

$$D\left(\frac{1}{v_{ref}} - \frac{1}{v_s}\right) = -\frac{1}{v_s}u_s \quad (277)$$

hence:

$$\frac{D - u_s}{v_s} = \frac{D}{v_{ref}} \quad (278)$$

momentum equation

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (279)$$

Hugoniot:

$$\dot{m}(u_s - u_0) = p_s - p_{ref} \quad (280)$$

$$\dot{m}u_s = p_s \quad (281)$$

energy equation

The hydro equations are:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (282)$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (283)$$

$$\frac{\partial e}{\partial t} + p \frac{\partial v}{\partial t} = 0 \quad (284)$$

Looking at the energy equation:

$$\frac{\partial e}{\partial t} + p \frac{\partial v}{\partial t} = 0 \quad (285)$$

plugging the continuity equation:

$$\frac{\partial e}{\partial t} + p \frac{\partial u}{\partial m} = 0 \quad (286)$$

$$\frac{\partial e}{\partial t} + \frac{\partial(pu)}{\partial m} - u \frac{\partial p}{\partial m} = 0 \quad (287)$$

plugging the momentum equation:

$$\frac{\partial e}{\partial t} + \frac{\partial(pu)}{\partial m} + u \frac{\partial u}{\partial t} = 0 \quad (288)$$

$$\frac{\partial e}{\partial t} + \frac{\partial (pu)}{\partial m} + \frac{\partial}{\partial t} \left(\frac{1}{2} u^2 \right) = 0 \quad (289)$$

and we get the **total energy** equation:

$$\frac{\partial}{\partial t} \left(e + \frac{1}{2} u^2 \right) + \frac{\partial (pu)}{\partial m} = 0 \quad (290)$$

Hugoniot:

$$\dot{m} \left(\left(e_s + \frac{1}{2} u_s^2 \right) - \left(e_{ref} + \frac{1}{2} u_{ref}^2 \right) \right) = u_s p_s - u_{ref} p_{ref} \quad (291)$$

$$\dot{m} \left(e_s + \frac{1}{2} u_s^2 \right) = u_s p_s \quad (292)$$

$$pv = (\gamma - 1) e \quad (293)$$

$$e = \frac{pv}{r} \quad (294)$$

$$\dot{m} \left(\frac{p_s v_s}{r} + \frac{1}{2} u_s^2 \right) = u_s p_s \quad (295)$$

summary,

Final dimensional jump conditions:

$$\dot{m} (v_s - v_{ref}) = -u_s \quad (296)$$

$$\dot{m} u_s = p_s \quad (297)$$

$$\dot{m} \left(\frac{p_s v_s}{r} + \frac{1}{2} u_s^2 \right) = u_s p_s \quad (298)$$

if one defined shock velocity D :

$$\dot{m} = \rho_{ref} D = \frac{D}{v_{ref}} \quad (299)$$

we get for the continuity eq:

$$\frac{D}{v_0} (v_s - v_{ref}) = -u_s \quad (300)$$

which is:

$$\frac{D - u_s}{v_s} = \frac{D}{v_{ref}} \quad (301)$$

momentum:

$$\frac{D}{v_{ref}} u_s = p_s \quad (302)$$

energy:

$$D \left(e_s + \frac{1}{2} u_s^2 \right) = v_{ref} u_s p_s \quad (303)$$

which is same as S-H.

6 Dimensionless jump conditions

now since

$$\xi = m v_0^{\frac{1}{2-\omega}} p_0^{-\frac{1-\omega}{2-\omega}} t^{-(2+\tau)\frac{1-\omega}{2-\omega}} \quad (304)$$

$$m = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi \quad (305)$$

$$\dot{m} = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi \quad (306)$$

we also had:

$$V = v v_0^{\frac{2}{\omega-2}} p_0^{\frac{\omega}{\omega-2}} t^{\frac{(\tau+2)\omega}{\omega-2}} \quad (307)$$

$$U = u v_0^{\frac{1}{\omega-2}} p_0^{\frac{1}{\omega-2}} t^{\frac{\omega+\tau}{\omega-2}} \quad (308)$$

$$P = \frac{p}{p_0 t^\tau} \quad (309)$$

so we will use the following relations:

$$m = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi \quad (310)$$

$$\dot{m} = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi \quad (311)$$

$$v = V v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (312)$$

$$u = U v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (313)$$

$$p = P p_0 t^\tau \quad (314)$$

continuity

$$\dot{m} (v_s - v_{ref}) = -u_s \quad (315)$$

the initial density profile is:

$$v(m, 0) = v_0^{\frac{1}{1-\omega}} [(1-\omega)m]^{\frac{\omega}{1-\omega}} \quad (316)$$

the initial density at the shock is:

$$v_{ref} \equiv v(m_s, 0) = v_0^{\frac{1}{1-\omega}} [(1-\omega)m_s]^{\frac{\omega}{1-\omega}} \quad (317)$$

$$= v_0^{\frac{1}{1-\omega}} \left[(1-\omega) v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi_s \right]^{\frac{\omega}{1-\omega}} \quad (318)$$

$$= v_0^{\frac{1}{1-\omega}} (1-\omega)^{\frac{\omega}{1-\omega}} v_0^{-\frac{\omega}{(2-\omega)(1-\omega)}} p_0^{\frac{\omega}{2-\omega}} t^{(2+\tau)\frac{\omega}{2-\omega}} \xi_s^{\frac{\omega}{1-\omega}} \quad (319)$$

$$\frac{1}{1-\omega} - \frac{\omega}{(2-\omega)(1-\omega)} = \frac{1}{1-\omega} \left[1 - \frac{\omega}{2-\omega} \right] = \frac{1}{1-\omega} \left[\frac{2-2\omega}{2-\omega} \right] = \frac{2}{2-\omega} \quad (320)$$

$$v_{ref} = v_0^{\frac{2}{2-\omega}} (1-\omega)^{\frac{\omega}{1-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{(2+\tau)\frac{\omega}{2-\omega}} \xi_s^{\frac{\omega}{1-\omega}} \quad (321)$$

check that it has units of inverse:

$$\left[v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{(2+\tau)\frac{\omega}{2-\omega}} \right] = [M^{-1} L^{1-\omega}]^{\frac{2}{2-\omega}} [MLT^{-\tau-2}]^{\frac{\omega}{2-\omega}} [T]^{(2+\tau)\frac{\omega}{2-\omega}} \quad (322)$$

$$= [M]^{\frac{-2}{2-\omega} + \frac{\omega}{2-\omega}} [L]^{\frac{2(1-\omega)}{2-\omega} + \frac{\omega}{2-\omega}} [T]^{\frac{\omega}{2-\omega}(-\tau-2) + (2+\tau)\frac{\omega}{2-\omega}} \quad (323)$$

$$= [M]^{-1} [L] \quad (324)$$

hence:

$$\dot{m}(v_s - v_{ref}) = -u_s \quad (325)$$

$$v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s \left(V_s v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} - v_0^{\frac{2}{2-\omega}} (1-\omega)^{\frac{\omega}{1-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{(2+\tau)\frac{\omega}{2-\omega}} \xi_s^{\frac{\omega}{1-\omega}} \right) = -U \quad (326)$$

temporal power cancels:

$$(\tau+2) \frac{1-\omega}{2-\omega} - 1 + \frac{(\tau+2)\omega}{2-\omega} - \frac{\omega+\tau}{2-\omega} = \frac{(\tau+2)(1-\omega+\omega)}{2-\omega} + \frac{-2+\omega-\omega-\tau}{2-\omega} \quad (327)$$

$$= \frac{\tau+2}{2-\omega} + \frac{-2-\tau}{2-\omega} = 0 \quad (328)$$

easy to see that v_0 and p_0 powers cancel as well.

Hence we get:

$$(2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s \left(V_s - ((1-\omega) \xi_s)^{\frac{\omega}{1-\omega}} \right) = -U_s \quad (329)$$

$$V_s - ((1-\omega) \xi_s)^{\frac{\omega}{1-\omega}} = \frac{-U_s}{(2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s} \quad (330)$$

$$V_s = ((1-\omega) \xi_s)^{\frac{\omega}{1-\omega}} - \frac{U_s}{(2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s} \quad (331)$$

which

is same as S-H.

momentum:

$$\dot{m}u_s = p_s \quad (332)$$

$$\left[v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi \right] u_s = p_s \quad (333)$$

$$\cancel{v_0^{-\frac{1}{2-\omega}}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s U_s \cancel{v_0^{\frac{1}{2-\omega}}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} = P_s p_0 t^\tau \quad (334)$$

$$p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s U_s p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} = P_s p_0 t^\tau \quad (335)$$

total p_0 power is:

$$\frac{1-\omega}{2-\omega} + \frac{1}{2-\omega} - 1 = 0 \quad (336)$$

total t power is:

$$(\tau+2) \frac{1-\omega}{2-\omega} - 1 + \frac{\omega+\tau}{2-\omega} - \tau = \quad (337)$$

$$\tau \left[\cancel{\frac{1-\omega}{2-\omega}} + \cancel{\frac{1}{2-\omega} - 1} \right] + \cancel{\frac{2-2\omega+\omega}{2-\omega} - 1} = 0 \quad (338)$$

hence:

$$(2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s U_s = P_s \quad (339)$$

$$P_s = (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s U_s \quad (340)$$

same as S-H.

energy

$$\dot{m} \left(\frac{p_s v_s}{r} + \frac{1}{2} u_s^2 \right) = u_s p_s \quad (341)$$

$$v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi \left(\frac{p_s v_s}{r} + \frac{1}{2} u_s^2 \right) = u_s p_s \quad (342)$$

$$v = V v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (343)$$

$$u = U v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (344)$$

$$p = P p_0 t^\tau \quad (345)$$

$$v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(\tau+2)\frac{1-\omega}{2-\omega}-1} (2+\tau) \left(\frac{1-\omega}{2-\omega} \right) \xi_s \left[\frac{1}{r} P_s p_0 t^\tau V_s v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} + \frac{1}{2} U_s^2 v_0^{\frac{2}{2-\omega}} p_0^{\frac{2}{2-\omega}} t^{\frac{2(\omega+\tau)}{2-\omega}} \right] = U_s v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} \quad (346)$$

easy to see now that, v_0 cancels on both sides of eq. 346.

p_0 power inside $[]$ parenthesis, equals on both terms:

$$1 + \frac{\omega}{2 - \omega} = \frac{2 - \omega + \omega}{2 - \omega} = \frac{2}{2 - \omega} \quad (347)$$

t power inside $[]$ parenthesis, equals on both terms:

$$\tau + \frac{(\tau + 2)\omega}{2 - \omega} = \frac{\tau(2 - \omega) + (\tau + 2)\omega}{2 - \omega} = \frac{2\omega + 2\tau}{2 - \omega} = \frac{2(\omega + \tau)}{2 - \omega} \quad (348)$$

total p_0 power on the lhs:

$$\frac{1 - \omega}{2 - \omega} + \frac{2}{2 - \omega} = \frac{2 - \omega}{2 - \omega} + \frac{1}{2 - \omega} = 1 + \frac{1}{2 - \omega} \quad (349)$$

which is exactly the p_0 power on the rhs.

total temporal power on the lhs, minus the power on the rhs:

$$(\tau + 2) \frac{1 - \omega}{2 - \omega} - 1 + \frac{2(\omega + \tau)}{2 - \omega} - \frac{\omega + \tau}{2 - \omega} - \tau = \quad (350)$$

$$\frac{(\tau + 2)(1 - \omega) - 2 + \omega + 2\omega + 2\tau - \omega - \tau - 2\tau + \tau\omega}{2 - \omega} \quad (351)$$

$$= \frac{(\tau + 2)(1 - \omega) - 2 + 2\omega - \tau + \tau\omega}{2 - \omega} \quad (352)$$

$$= \frac{\tau - \tau\omega + 2 - 2\omega - 2 + 2\omega - \tau + \tau\omega}{2 - \omega} = 0 \quad (353)$$

hence, all units cancel and we get:

$$(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s \left(\frac{P_s V_s}{r} + \frac{1}{2} U_s^2 \right) = U_s P_s \quad (354)$$

same as S-H.

summary,

dimensionless jump conditions:

$$V_s = ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} - \frac{U_s}{(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s} \quad (355)$$

$$P_s = (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s U_s \quad (356)$$

$$(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s \left(\frac{P_s V_s}{r} + \frac{1}{2} U_s^2 \right) = U_s P_s \quad (357)$$

all are same as S-H eq. 45

let solve this system of three equations in three variables.

Putting the second eq. in the third:

$$(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s \left(\frac{V_s}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s U_s + \frac{1}{2} U_s^2 \right) = U_s (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s U_s \quad (358)$$

$$\frac{V_s}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s + \frac{1}{2} U_s = U_s \quad (359)$$

$$U_s = \frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \quad (360)$$

put this result in the first eq.:

$$V_s = ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} - \frac{U_s}{(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s} \quad (361)$$

$$V_s = ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} - \frac{\frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s}{(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s} \quad (362)$$

$$V_s = ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} - \frac{2}{r} V_s \quad (363)$$

$$V_s \left(1 + \frac{2}{r} \right) = ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (364)$$

$$V_s = \frac{((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}}}{1 + \frac{2}{r}} \quad (365)$$

$$V_s = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (366)$$

$$V_s = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (367)$$

$$V_s = \frac{\gamma - 1}{\gamma + 1} ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (368)$$

Now

$$P_s = (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s U_s \quad (369)$$

since

$$U_s = \frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \quad (370)$$

$$(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s = \frac{r}{2} \frac{U_s}{V_s} \quad (371)$$

so that

$$P_s = \frac{r}{2} \frac{U_s}{V_s} U_s \quad (372)$$

$$P_s = \frac{U_s^2}{2} \frac{r}{V_s} \quad (373)$$

summary

Final dimensionless jump conditions:

$$V_s = \frac{\gamma - 1}{\gamma + 1} ((1 - \omega) \xi_s)^{\frac{\omega}{1-\omega}} = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1-\omega}} \quad (374)$$

$$U_s = \frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \quad (375)$$

$$P_s = \frac{U_s^2}{2} \frac{r}{V_s} \quad (376)$$

all are same as S-H eq. 46

7 Relation to Eulerian coordinate - positions in space

The shocked mass is:

$$m_s(t) = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi_s \quad (377)$$

which gives a condition

$$(2 + \tau) \frac{1 - \omega}{2 - \omega} > 0 \quad (378)$$

and since $\omega < 1$:

$$\tau > -2 \quad (379)$$

we have the relations:

$$m(t, x) = \int_0^x \rho(x', t) dx' \quad (380)$$

$$x(t, m) - x_p(t) = \int_0^m \frac{dm'}{\rho(m', t)} \quad (381)$$

where $x_p(t)$ is the piston position (the system boundary on which the external pressure is applied).

7.1 Exact positions for given mass

distance from piston as a function of mass

$$x(t, m) - x_p(t) = \int_0^m \frac{dm'}{\rho(m', t)} = \int_0^m v(m', t) dm' \quad (382)$$

$$= \int_0^m V(\xi') v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} dm' \quad (383)$$

$$m' = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi' \quad (384)$$

$$dm' = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} d\xi' \quad (385)$$

$$x(t, m) - x_p(t) = \int_0^{\xi(m)} V(\xi') v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} d\xi' \quad (386)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \int_0^{\xi(m)} V(\xi') d\xi' \quad (387)$$

$$(2+\tau) \frac{1-\omega}{2-\omega} + \frac{(\tau+2)\omega}{2-\omega} = \frac{2-2\omega+\tau-\tau\omega+\tau\omega+2\omega}{2-\omega} = \frac{2+\tau}{2-\omega} \quad (388)$$

$$x(t, m) - x_p(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2+\tau}{2-\omega}} \int_0^{\xi(m)} V(\xi') d\xi' \quad (389)$$

$$x(t, m) - x_p(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \int_0^{\xi(m)} V(\xi') d\xi' \quad (390)$$

7.2 Distance from piston to shock

The distance from piston (the system boundary on which the external pressure is applied) to shock is:

$$\Delta x_s(t) = x(t, m_s) - x_p(t) \quad (391)$$

$$\Delta x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \int_0^{\xi_s} V(\xi') d\xi' \quad (392)$$

so we need to calculate the area under $V(\xi)$.

7.3 Piston position

$$u_p(t) \equiv u(m=0, t) \quad (393)$$

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (394)$$

$$u_p(t) \equiv U(0) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (395)$$

$$x_p(t) = \int_0^t u_p(t') dt' = \int_0^t U(0) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t'^{\frac{\omega+\tau}{2-\omega}} dt' \quad (396)$$

$$= U(0) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} \int_0^t t'^{\frac{\omega+\tau}{2-\omega}} dt' \quad (397)$$

$$= U(0) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} \frac{t^{\frac{\omega+\tau}{2-\omega}+1}}{\frac{\omega+\tau}{2-\omega}+1} \quad (398)$$

$$= U(0) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\tau+2}{2-\omega}} \left(\frac{2-\omega}{\tau+2} \right) \quad (399)$$

hence the piston position is:

$$x_p(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} U(0) \left(\frac{2-\omega}{\tau+2} \right) \quad (400)$$

condition on τ due to the temporal integration?

$$\frac{\omega+\tau}{2-\omega} > -1 \quad (401)$$

$$\omega+\tau > -2+\omega \quad (402)$$

$$\tau > -2 \quad (403)$$

which we had already (condition that shock moves outwards).

The pi

7.4 Position from temporal integration of velocity?

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (404)$$

$$x(m, t) - x(m, 0) = \int_0^t u(m, t') dt' = \int_0^t U(\xi(m, t')) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t'^{\frac{\omega+\tau}{2-\omega}} dt' \quad (405)$$

$$m = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi \quad (406)$$

$$t = \left[\frac{m}{\xi} v_0^{\frac{1}{2-\omega}} p_0^{-\frac{1-\omega}{2-\omega}} \right]^{\frac{2-\omega}{(2+\tau)(1-\omega)}} \quad (407)$$

$$t = \left(\frac{m}{\xi} \right)^{\frac{2-\omega}{(2+\tau)(1-\omega)}} v_0^{\frac{1}{(2+\tau)(1-\omega)}} p_0^{-\frac{1}{(2+\tau)}} \quad (408)$$

$$t = \left(\frac{\xi}{m} \right)^{-\frac{2-\omega}{(2+\tau)(1-\omega)}} v_0^{\frac{1}{(2+\tau)(1-\omega)}} p_0^{-\frac{1}{(2+\tau)}} \quad (409)$$

$$dt = \left(-\frac{2-\omega}{(2+\tau)(1-\omega)} \right) \left(\frac{\xi}{m} \right)^{-\frac{2-\omega}{(2+\tau)(1-\omega)}-1} v_0^{\frac{1}{(2+\tau)(1-\omega)}} p_0^{-\frac{1}{2+\tau}} d\xi \quad (410)$$

$$x(m, t) - x(m, 0) = \int_0^t U(\xi(m, t')) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t'^{\frac{\omega+\tau}{2-\omega}} dt' \quad (411)$$

$$= \int_0^{\xi(m, t)} U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} \left[\left(\frac{\xi}{m} \right)^{-\frac{2-\omega}{(2+\tau)(1-\omega)}} v_0^{\frac{1}{(2+\tau)(1-\omega)}} p_0^{-\frac{1}{(2+\tau)}} \right]^{\frac{\omega+\tau}{2-\omega}} \left(-\frac{2-\omega}{(2+\tau)(1-\omega)} \right) \left(\frac{\xi}{m} \right)^{\frac{\omega+\tau}{2-\omega}} d\xi \quad (412)$$

7.5 Shock position

$$x_s(t) \equiv x_p(t) + \Delta x_s(t) \quad (413)$$

$$x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} U(0) \left(\frac{2-\omega}{\tau+2} \right) + (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \int_0^{\xi_s} V(\xi') d\xi' \quad (414)$$

$$x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[U(0) \left(\frac{2-\omega}{\tau+2} \right) + \int_0^{\xi_s} V(\xi') d\xi' \right] \quad (415)$$

7.6 Shock position from initial density

We can also calculate the shock position using the initial (unperturbed) position of the shocked mass (since the position there is exactly unperturbed).

the position of mass m for which its radius r is unperturbed is eq. 251:

$$r = [v_0 (1-\omega) m]^{\frac{1}{1-\omega}} \quad (416)$$

hence:

$$x_s(t) = [v_0 (1-\omega) m_s(t)]^{\frac{1}{1-\omega}} \quad (417)$$

we saw that the shocked mass is:

$$m_s(t) = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi_s \quad (418)$$

$$x_s(t) = \left[v_0 (1-\omega) v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi_s \right]^{\frac{1}{1-\omega}} \quad (419)$$

$$= v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2+\tau}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (420)$$

hence:

$$x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (421)$$

BUT on the other hand we saw

$$x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[U(0) \left(\frac{2-\omega}{\tau+2} \right) + \int_0^{\xi_s} V(\xi') d\xi' \right] \quad (422)$$

hence we must have:

$$\int_0^{\xi_s} V(\xi') d\xi' = ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} - U(0) \left(\frac{2-\omega}{\tau+2} \right) \quad (423)$$

we will see below in section 8, that this can be obtained exactly using the Jump conditions and an integral identity for V .

7.7 Shock velocity

$$x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (424)$$

$$v_s(t) = \frac{dx_s}{dt} = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (425)$$

$$= \frac{2+\tau}{2-\omega} \frac{1}{t} (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (426)$$

$$= \frac{2+\tau}{2-\omega} \frac{1}{t} x_s(t) \quad (427)$$

7.8 Time for shock position to reach a given position

time for shock to reach an position x :

$$x = (v_0 p_0 t_s(x)^{2+\tau})^{\frac{1}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (428)$$

$$\left(\frac{x}{((1-\omega) \xi_s)^{\frac{1}{1-\omega}}} \right)^{2-\omega} = v_0 p_0 t_s(x)^{2+\tau} \quad (429)$$

$$t_s(x) = \left[\frac{1}{v_0 p_0} \left(\frac{x}{((1-\omega) \xi_s)^{\frac{1}{1-\omega}}} \right)^{2-\omega} \right]^{\frac{1}{2+\tau}} \quad (430)$$

7.9 Positions as a function of time

For any mass coordinate m , In unshocked region:

$$x(m, t) = [v_0 (1-\omega) m]^{\frac{1}{1-\omega}} \quad (431)$$

In shocked region:

$$x(t, m) - x_p(t) = \int_0^m \frac{dm'}{\rho(m', t)} = \int_0^m v(m', t) dm' \quad (432)$$

Hence:

for the shocked region:

$$\xi(m) = m v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (433)$$

$$x(t, m) = x_p(t) + (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \int_0^{\xi(m)} V(\xi') d\xi' \quad (434)$$

$$x(t, m) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[U(0) \left(\frac{2-\omega}{\tau+2} \right) + \int_0^{\xi(m)} V(\xi') d\xi' \right] \quad (435)$$

in the unshocked region:

$$r(t, m) = [v_0 (1-\omega) m]^{\frac{1}{1-\omega}} \quad (436)$$

these two expressions coincide on the shock (will be proved in sec. 8):

$$[v_0 (1-\omega) m_s(t)]^{\frac{1}{1-\omega}} = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[U(0) \left(\frac{2-\omega}{\tau+2} \right) + \int_0^{\xi_s} V(\xi') d\xi' \right] \quad (437)$$

8 Calculation of $V(\xi)$ integral

the self similar continuity equation is:

$$\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1) \xi V' + \omega V) - U' = 0 \quad (438)$$

$$(\omega-1) \xi V' + \omega V = \frac{2-\omega}{\tau+2} U' \quad (439)$$

$$V(\xi) = \left(\frac{1}{\omega} - 1 \right) \xi V' + \frac{2-\omega}{\omega(\tau+2)} U' \quad (440)$$

$$V(\xi) = p \xi V' + q U' \quad (441)$$

$$p = \frac{1}{\omega} - 1 \quad (442)$$

$$q = \frac{2-\omega}{\omega(\tau+2)} \quad (443)$$

integration

$$\int_0^{\xi_0} V(\xi) d\xi = p \int_0^{\xi_0} \xi V'(\xi) d\xi + q (U(\xi_0) - U(0)) \quad (444)$$

integration by parts:

$$\int_0^{\xi_0} \xi V'(\xi) d\xi = [\xi V(\xi)]_0^{\xi_0} - \int_0^{\xi_0} V(\xi) d\xi \quad (445)$$

hence:

$$\int_0^{\xi_0} V(\xi) d\xi = p \left[[\xi V(\xi)]_0^{\xi_0} - \int_0^{\xi_0} V(\xi) d\xi \right] + q (U(\xi_0) - U(0)) \quad (446)$$

$$(1+p) \int_0^{\xi_0} V(\xi) d\xi = p [\xi V(\xi)]_0^{\xi_0} + q (U(\xi_0) - U(0)) \quad (447)$$

$$\int_0^{\xi_0} V(\xi) d\xi = \frac{p}{1+p} [\xi V(\xi)]_0^{\xi_0} + \frac{q}{1+p} (U(\xi_0) - U(0)) \quad (448)$$

$$\frac{p}{1+p} = \frac{\frac{1}{\omega} - 1}{1 + \frac{1}{\omega} - 1} = \frac{\frac{1}{\omega} - 1}{\frac{1}{\omega}} = 1 - \omega \quad (449)$$

$$\frac{q}{1+p} = \frac{\frac{2-\omega}{\omega(\tau+2)}}{1 + \frac{1}{\omega} - 1} = \frac{\frac{2-\omega}{\omega(\tau+2)}}{\frac{1}{\omega}} = \frac{2-\omega}{\tau+2} \quad (450)$$

hence:

$$\int_0^{\xi_0} V(\xi) d\xi = q_1 [\xi V(\xi)]_0^{\xi_0} + q_2 (U(\xi_0) - U(0)) \quad (451)$$

$$q_1 = 1 - \omega, \quad q_2 = \frac{2 - \omega}{\tau + 2} \quad (452)$$

what is

$$\lim_{\xi \rightarrow 0} [V(\xi) \xi] \quad (453)$$

expand near zero

$$\lim_{\xi \rightarrow 0} V(\xi) \approx A \xi^k \quad (454)$$

since $\int_0^{\xi_0} V(\xi) d\xi$ represents the position of mass $m(\xi_0)$, it must be finite (for any m and hence ξ_0):

$$\int_0^{\xi_0} V(\xi) d\xi = \int_0^{\xi_0} A \xi^k d\xi \quad (455)$$

this integral is finite only if

$$k > -1 \quad (456)$$

or

$$k + 1 > 0 \quad (457)$$

hence

$$\lim_{\xi \rightarrow 0} [V(\xi) \xi] \approx \lim_{\xi \rightarrow 0} [A \xi^{k+1}] = 0 \quad (458)$$

hence:

$$\int_0^{\xi_0} V(\xi) d\xi = q_1 \xi_0 V(\xi_0) + q_2 (U(\xi_0) - U(0)) \quad (459)$$

Summary

for any $\xi \leq \xi_s$:

$$\int_0^\xi V(\xi') d\xi' = q_1 \xi V(\xi) + q_2 (U(\xi) - U(0)) \quad (460)$$

$$q_1 = 1 - \omega, \quad q_2 = \frac{2 - \omega}{\tau + 2} \quad (461)$$

specifically for $\xi_0 = \xi_s$:

$$\int_0^{\xi_s} V(\xi') d\xi' = q_1 \xi_s V_s + q_2 (U_s - U(0)) \quad (462)$$

Hugoniot:

$$V_s = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (463)$$

$$\int_0^{\xi_s} V(\xi') d\xi' = q_1 \xi_s V_s + q_2 (U_s - U(0)) \quad (464)$$

$$= (1 - \omega) \xi_s V_s + \frac{2 - \omega}{\tau + 2} (U_s - U(0)) \quad (465)$$

$$= (1 - \omega) \xi_s V_s + \frac{2 - \omega}{\tau + 2} U_s - \frac{2 - \omega}{\tau + 2} U(0) \quad (466)$$

Jump conditions:

$$U_s = \frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \quad (467)$$

$$\int_0^{\xi_s} V(\xi') d\xi' = (1 - \omega) \xi_s V_s + \frac{2 - \omega}{\tau + 2} U_s - \frac{2 - \omega}{\tau + 2} U(0) \quad (468)$$

$$= (1 - \omega) \xi_s V_s + \frac{2 - \omega}{\tau + 2} \left[\frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \right] - \frac{2 - \omega}{\tau + 2} U(0) \quad (469)$$

$$= (1 - \omega) \xi_s V_s + \left[\frac{2}{r} (1 - \omega) \xi_s V_s \right] - \frac{2 - \omega}{\tau + 2} U(0) \quad (470)$$

$$= (1 - \omega) \xi_s V_s \left(1 + \frac{2}{r} \right) - \frac{2 - \omega}{\tau + 2} U(0) \quad (471)$$

$$= (1 - \omega) \xi_s V_s \left(\frac{r + 2}{r} \right) - \frac{2 - \omega}{\tau + 2} U(0) \quad (472)$$

Jump conditions:

$$V_s = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (473)$$

hence:

$$\int_0^{\xi_s} V(\xi') d\xi' = (1 - \omega) \xi_s V_s \left(\frac{r+2}{r} \right) - \frac{2-\omega}{\tau+2} U(0) \quad (474)$$

$$= (1 - \omega) \xi_s \left[\frac{r}{r+2} ((1 - \omega) \xi_s)^{\frac{1-\omega}{1-\omega}} \right] \left(\frac{r+2}{r} \right) - \frac{2-\omega}{\tau+2} U(0) \quad (475)$$

$$= ((1 - \omega) \xi_s)^{\frac{1}{1-\omega}} - \frac{2-\omega}{\tau+2} U(0) \quad (476)$$

In summary, from our integral identity we found that:

$$\int_0^{\xi_s} V(\xi') d\xi' = ((1 - \omega) \xi_s)^{\frac{1}{1-\omega}} - U(0) \left(\frac{2-\omega}{\tau+2} \right) \quad (477)$$

WHICH IS EXACTLY WHAT WE ALREADY SAW from the calculation of shock position in eq. 421-423 due to the shock position from initial density and shock mass.

So this is a very nice check.

We can finally write nicely the position of a Lagrangian element:

$$x(t, m) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[U(0) \left(\frac{2-\omega}{\tau+2} \right) + \int_0^{\xi(m)} V(\xi') d\xi' \right] \quad (478)$$

$$x(t, m) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[U(0) \left(\frac{2-\omega}{\tau+2} \right) + \left((1 - \omega) \xi V(\xi) + \left(\frac{2-\omega}{\tau+2} \right) (U(\xi) - U(0)) \right) \right] \quad (479)$$

$$x(t, m) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[(1 - \omega) \xi V(\xi) + \left(\frac{2-\omega}{\tau+2} \right) U(\xi) \right] \quad (480)$$

9 Energy

internal energy:

$$E_{in}(t) = \int_0^{m_s(t)} e(m, t) dm = \frac{1}{r} \int_0^{m_s(t)} p(m, t) v(m, t) dm \quad (481)$$

$$v(m, t) = V(\xi) v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (482)$$

$$p(m, t) = P(\xi) p_0 t^\tau \quad (483)$$

$$m = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \xi \quad (484)$$

$$dm = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} d\xi \quad (485)$$

$$E_{in}(t) = \frac{1}{r} \int_0^{m_s(t)} p(m, t) v(m, t) dm \quad (486)$$

$$= \frac{1}{r} \int_0^{\xi_s} (P(\xi) p_0 t^\tau) \left(V(\xi) v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \right) \left(v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \right) d\xi \quad (487)$$

p_0 power:

$$\frac{\omega}{2-\omega} + \frac{1-\omega}{2-\omega} + 1 = \frac{3-\omega}{2-\omega} \quad (488)$$

t power:

$$\tau + \frac{(\tau+2)\omega}{2-\omega} + (2+\tau) \frac{1-\omega}{2-\omega} = \tau + \frac{\tau+2}{2-\omega} = \frac{3\tau - \tau\omega + 2}{2-\omega} \quad (489)$$

$$E_{in}(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau - \tau\omega + 2}{2-\omega}} \int_0^{\xi_s} \frac{P(\xi) V(\xi)}{r} d\xi \quad (490)$$

p	u	v	v_0	p_0	m	t
MLT^{-2}	LT^{-1}	$M^{-1}L$	$M^{-1}L^{1-\omega}$	$MLT^{-\tau-2}$	M	T

units check should be $[ML^2/T^2]$:

mass:

$$-\frac{1}{2-\omega} + \frac{3-\omega}{2-\omega} = 1 \quad (491)$$

length:

$$\frac{1-\omega}{2-\omega} + \frac{3-\omega}{2-\omega} = 2 \quad (492)$$

temporal power:

$$(-\tau-2) \frac{3-\omega}{2-\omega} + \frac{3\tau - \tau\omega + 2}{2-\omega} = \frac{-3\tau + \tau\omega - 6 + 2\omega + 3\tau - \tau\omega + 2}{2-\omega} \quad (493)$$

$$= \frac{-4 + 2\omega}{2-\omega} = -2 \quad (494)$$

so the internal energy:

$$E_{in}(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau - \tau\omega + 2}{2-\omega}} \int_0^{\xi_s} \frac{P(\xi) V(\xi)}{r} d\xi \quad (495)$$

Kinetic energy:

$$E_k(t) = \int_0^{m_s(t)} \frac{1}{2} u^2(m, t) dm \quad (496)$$

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (497)$$

$$E_k(t) = \frac{1}{2} \int_0^{\xi_s} \left(U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \right)^2 \left(v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{(2+\tau)\frac{1-\omega}{2-\omega}} \right) d\xi \quad (498)$$

p_0 power:

$$\frac{2}{2-\omega} + \frac{1-\omega}{2-\omega} = \frac{3-\omega}{2-\omega} \quad (499)$$

t power:

$$2\frac{\omega+\tau}{2-\omega} + (2+\tau)\frac{1-\omega}{2-\omega} = \frac{2\omega+2\tau+2-2\omega+\tau-\tau\omega}{2-\omega} = \frac{3\tau-\tau\omega+2}{2-\omega} \quad (500)$$

so the kinetic energy:

$$E_k(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \frac{U^2(\xi)}{2} d\xi \quad (501)$$

summary:

internal energy:

$$E_{in}(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \frac{P(\xi)V(\xi)}{r} d\xi \quad (502)$$

kinetic energy:

$$E_k(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \frac{U^2(\xi)}{2} d\xi \quad (503)$$

we have the identity:

$$\frac{E_k(t)}{E_{in}(t)} = \text{const} \quad (504)$$

total energy:

$$E(t) = E_{in}(t) + E_k(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \left(\frac{P(\xi)V(\xi)}{r} + \frac{U^2(\xi)}{2} \right) d\xi \quad (505)$$

Trying to get exact expression for integral.

Total energy equation:

$$\frac{\partial}{\partial t} \left(e + \frac{1}{2} u^2 \right) + \frac{\partial (pu)}{\partial m} = 0 \quad (506)$$

$$\frac{\partial}{\partial t} \left(\frac{pv}{r} + \frac{1}{2} u^2 \right) + \frac{\partial (pu)}{\partial m} = 0 \quad (507)$$

we already saw:

$$\frac{\partial v}{\partial t} = v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} \left(\frac{\tau+2}{2-\omega} \right) ((\omega-1)\xi V' + \omega V) \quad (508)$$

$$\frac{\partial p}{\partial t} = p_0 t^{\tau-1} \left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) \quad (509)$$

$$\frac{\partial u}{\partial t} = v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2\omega+\tau-2}{2-\omega}} \left(\xi U' \frac{(\tau+2)(\omega-1)}{2-\omega} + U \left(\frac{\omega+\tau}{2-\omega} \right) \right) \quad (510)$$

$$\frac{\partial p}{\partial m} = v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{2\omega+\tau-2}{2-\omega}} P' \quad (511)$$

$$\frac{\partial u}{\partial m} = v_0^{\frac{2}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{\tau\omega+3\omega-2}{2-\omega}} U' \quad (512)$$

assuming all units drop:

$$\frac{1}{r} \left[\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1) \xi V' + \omega V) P + \left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) V \right] + \left(\xi U' \frac{(\tau+2)(\omega-1)}{2-\omega} + U \left(\frac{\omega+\tau}{2-\omega} \right) \right) U + \quad (513)$$

$$\frac{1}{r} \left[\left(\frac{\tau+2}{2-\omega} \right) ((\omega-1) \xi V' + \omega V) P + \left(\frac{(\tau+2)(\omega-1)}{2-\omega} P' \xi + \tau P \right) V \right] + \left(\xi U' \frac{(\tau+2)(\omega-1)}{2-\omega} + U \left(\frac{\omega+\tau}{2-\omega} \right) \right) U + \quad (514)$$

the PV coefficient is:

$$\frac{1}{r} \left[\left(\frac{\tau+2}{2-\omega} \right) \omega + \tau \right] = \frac{1}{r} \left[\frac{\tau\omega + 2\omega + 2\tau - \tau\omega}{2-\omega} \right] = \frac{1}{r} \left[\frac{2\omega + 2\tau}{2-\omega} \right] = \frac{2}{r} \left(\frac{\omega+\tau}{2-\omega} \right) \quad (515)$$

hence:

$$2 \left(\frac{\omega+\tau}{2-\omega} \right) \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) = -\frac{1}{r} \left[\left(\frac{\tau+2}{2-\omega} \right) (\omega-1) \xi V' P + \frac{(\tau+2)(\omega-1)}{2-\omega} \xi P' V \right] - \frac{(\tau+2)(\omega-1)}{2-\omega} \xi U U' - (PU)' \quad (516)$$

$$2 \left(\frac{\omega+\tau}{2-\omega} \right) \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) = -\frac{1}{r} \left(\frac{\tau+2}{2-\omega} \right) (\omega-1) \xi [V' P + P' V] - \frac{1}{2} \frac{(\tau+2)(\omega-1)}{2-\omega} \xi (U^2)' - (PU)' \quad (517)$$

$$2 \left(\frac{\omega+\tau}{2-\omega} \right) \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) = - \left(\frac{\tau+2}{2-\omega} \right) (\omega-1) \xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right)' - (PU)' \quad (518)$$

$$\frac{PV}{r} + \frac{1}{2} U^2 = -\frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (\omega-1) \xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right)' - \frac{1}{2} \left(\frac{2-\omega}{\omega+\tau} \right) (PU)' \quad (519)$$

$$\int_0^{\xi_0} \xi f'(\xi) d\xi = [\xi f(\xi)]_0^{\xi_0} - \int_0^{\xi_0} f(\xi) d\xi \quad (520)$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = -\frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (\omega-1) \left(\left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi \right) - \frac{1}{2} \left(\frac{2-\omega}{\omega+\tau} \right) [PU]_0^{\xi_0} \quad (521)$$

$$\left(1 - \frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (\omega-1) \right) \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = -\frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (\omega-1) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - \frac{1}{2} \left(\frac{2-\omega}{\omega+\tau} \right) [PU]_0^{\xi_0} \quad (522)$$

$$\left(1 + \frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (1-\omega) \right) \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = -\frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (\omega-1) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - \frac{1}{2} \left(\frac{2-\omega}{\omega+\tau} \right) [PU]_0^{\xi_0} \quad (523)$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \frac{-\frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (\omega-1) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - \frac{1}{2} \left(\frac{2-\omega}{\omega+\tau} \right) [PU]_0^{\xi_0}}{1 + \frac{1}{2} \left(\frac{\tau+2}{\omega+\tau} \right) (1-\omega)} \quad (524)$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \frac{\left(\frac{2+\tau}{\omega+\tau} \right) (1-\omega) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - \left(\frac{2-\omega}{\omega+\tau} \right) [PU]_0^{\xi_0}}{2 + \left(\frac{\tau+2}{\omega+\tau} \right) (1-\omega)} \quad (525)$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \frac{(2+\tau)(1-\omega) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - (2-\omega) [PU]_0^{\xi_0}}{2(\omega+\tau) + (\tau+2)(1-\omega)} \quad (526)$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \frac{(2+\tau)(1-\omega) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - (2-\omega) [PU]_0^{\xi_0}}{2\omega + 2\tau + \tau - \tau\omega + 2 - 2\omega} \quad (527)$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \frac{(2+\tau)(1-\omega) \left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - (2-\omega) [PU]_0^{\xi_0}}{3\tau - \tau\omega + 2} \quad (528)$$

$$[PU]_0^\xi = P(\xi)U(\xi) - U(0) \quad (529)$$

since $\xi V \rightarrow 0, \xi U \rightarrow 0$ at $\xi \rightarrow 0$, we get:

$$\left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^\xi = \xi \left(\frac{P(\xi)V(\xi)}{r} + \frac{1}{2} U^2(\xi) \right) \quad (530)$$

$$\boxed{\int_0^\xi \left(\frac{P(\xi')V(\xi')}{r} + \frac{1}{2} U^2(\xi') \right) d\xi' = \frac{(2+\tau)(1-\omega)}{3\tau - \tau\omega + 2} \xi \left(\frac{P(\xi)V(\xi)}{r} + \frac{1}{2} U^2(\xi) \right) - \frac{2-\omega}{3\tau - \tau\omega + 2} (P(\xi)U(\xi) - U(0))} \quad (531)$$

For the total integral on $\xi = \xi_s$:

$$\int_0^{\xi_s} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi' = \frac{(2+\tau)(1-\omega)}{3\tau - \tau\omega + 2} \xi_s \left(\frac{P_s V_s}{r} + \frac{1}{2} U_s^2 \right) - \frac{2-\omega}{3\tau - \tau\omega + 2} (P_s U_s - U(0)) \quad (532)$$

we had the Hugoniot relation:

$$(2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s \left(\frac{P_s V_s}{r} + \frac{1}{2} U_s^2 \right) = U_s P_s \quad (533)$$

so that:

$$\int_0^{\xi_s} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi' = \frac{2 - \omega}{3\tau - \tau\omega + 2} U_s P_s - \frac{2 - \omega}{3\tau - \tau\omega + 2} (P_s U_s - U(0)) \quad (534)$$

$$\int_0^{\xi_s} \left(\frac{P(\xi) V(\xi)}{r} + \frac{1}{2} U^2(\xi) \right) d\xi = \frac{2 - \omega}{3\tau - \tau\omega + 2} U(0) \quad (535)$$

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Trying to get exact expression for individual internal/kinetic integral:
mass equation:

$$\left(\frac{\tau + 2}{2 - \omega} \right) ((\omega - 1) \xi V' + \omega V) - U' = 0 \quad (536)$$

multiply by U :

$$\left(\frac{\tau + 2}{2 - \omega} \right) ((\omega - 1) \xi UV' + \omega UV) - UU' = 0 \quad (537)$$

$$\left(\frac{\tau + 2}{2 - \omega} \right) ((\omega - 1) \xi UV' + \omega UV) - \frac{1}{2} (U^2)' = 0 \quad (538)$$

$$(\omega - 1) \xi UV' + \omega UV - \frac{1}{2} \left(\frac{2 - \omega}{\tau + 2} \right) (U^2)' = 0 \quad (539)$$

momentum equation:

$$\frac{(\tau + 2)(\omega - 1)}{2 - \omega} \xi U' + U \left(\frac{\omega + \tau}{2 - \omega} \right) + P' = 0 \quad (540)$$

$$\frac{(\tau + 2)(\omega - 1)}{\omega + \tau} \xi U' + U + \left(\frac{2 - \omega}{\omega + \tau} \right) P' = 0 \quad (541)$$

multiply by ωV :

$$\frac{(\tau + 2)(\omega - 1)\omega}{\omega + \tau} \xi VU' + \omega VU + \left(\frac{2 - \omega}{\omega + \tau} \right) \omega VP' = 0 \quad (542)$$

subtracting momentum from mass equation the ωVU term cancels:

$$\left[(\omega - 1) - \frac{(\tau + 2)(\omega - 1)\omega}{\omega + \tau} \right] \xi UV' - \frac{1}{2} \left(\frac{2 - \omega}{\tau + 2} \right) (U^2)' - \left(\frac{2 - \omega}{\omega + \tau} \right) \omega VP' = 0 \quad (543)$$

$$\frac{1}{2} (U^2)' = a\xi UV' + bVP' \quad (544)$$

integrating:

$$\int_0^{\xi_0} \frac{U^2(\xi)}{2} d\xi = \int_0^{\xi_0} (a\xi UV' + bVP') d\xi \quad (545)$$

$$\int_0^{\xi_0} VP' d\xi = [VP]_0^{\xi_0} - \int_0^{\xi_0} V' P d\xi \quad (546)$$

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trying to find an equation to total kinetic energy

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (547)$$

$$u \frac{\partial u}{\partial t} + u \frac{\partial p}{\partial m} = 0 \quad (548)$$

$$\frac{\partial}{\partial t} \left(\frac{1}{2} u^2 \right) + u \frac{\partial p}{\partial m} = 0 \quad (549)$$

$$\frac{\partial}{\partial t} \left(\frac{1}{2} u^2 \right) + \frac{\partial (pu)}{\partial m} - p \frac{\partial u}{\partial m} = 0 \quad (550)$$

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10 Special case $\tau = \omega = 0$

In this case the solution is constant and we have

$$P_s = P(0) = 1 \quad (551)$$

so we can find ξ_s analytically from the hugoniot relations:

$$V_s = \frac{\gamma - 1}{\gamma + 1} ((1 - \omega) \xi_s)^{\frac{\omega}{1-\omega}} = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1-\omega}} \quad (552)$$

$$U_s = \frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \quad (553)$$

$$P_s = \frac{U_s^2}{2} \frac{r}{V_s} \quad (554)$$

hence (we can drop the s on the constant profiles):

$$V = \frac{r}{r + 2} \quad (555)$$

$$U = \frac{2}{r} \xi_s V \quad (556)$$

$$P = \frac{U^2}{2} \frac{r}{V} = 1 \quad (557)$$

from the last equation:

$$U = \left(\frac{2}{r} V \right)^{\frac{1}{2}} = \left(\frac{2}{r} \frac{r}{r+2} \right)^{\frac{1}{2}} = \left(\frac{2}{r+2} \right)^{\frac{1}{2}} \quad (558)$$

using this in the momentum eq:

$$U = \frac{2}{r} \xi_s V \quad (559)$$

$$\left(\frac{2}{r+2} \right)^{\frac{1}{2}} = \frac{2}{r} \xi_s \frac{r}{r+2} \quad (560)$$

$$\xi_s = \left(\frac{r+2}{2} \right)^{\frac{1}{2}} \quad (561)$$

summary:

$$V(\xi) = \frac{r}{r+2} \quad (562)$$

$$U(\xi) = \left(\frac{2}{r+2} \right)^{\frac{1}{2}} \quad (563)$$

$$P(\xi) = 1 \quad (564)$$

$$\xi_s = \left(\frac{r+2}{2} \right)^{\frac{1}{2}} \quad (565)$$

11 grand summary

hydro equations:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \quad (566)$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \quad (567)$$

$$\frac{\partial}{\partial t} \left(\frac{pv}{r} \right) + p \frac{\partial v}{\partial t} = 0 \quad (568)$$

$$r = \gamma - 1 \quad (569)$$

solved with a Lagrangian piston boundary condition:

$$p(m=0, t) = p_0 t^\tau \quad (570)$$

and with an initial Eulerian condition:

$$\rho(r, 0) = \rho_0 r^{-\omega} \implies v(r, 0) = v_0 r^\omega \quad (571)$$

in order to have a finite mass, we have:

$$\omega < 1 \quad (572)$$

which gives an initial Lag. condition:

$$v(m, 0) = v_0^{\frac{1}{1-\omega}} [(1-\omega)m]^{\frac{\omega}{1-\omega}} \quad (573)$$

$$u(m, 0) = 0, \quad p(m, 0) = 0 \quad (574)$$

The self similar representation of this problem:

$$\xi = m v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(\tau+2)(\omega-1)}{2-\omega}} \quad (575)$$

$$v(m, t) = V(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(\tau+2)\omega}{2-\omega}} \quad (576)$$

$$u(m, t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \quad (577)$$

$$p(m, t) = P(\xi) p_0 t^\tau \quad (578)$$

the self similar profiles obey the ODE:

$$\left(\frac{\tau+2}{2-\omega}\right) ((\omega-1)\xi V' + \omega V) - U' = 0 \quad (579)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \xi U' + U \left(\frac{\omega+\tau}{2-\omega}\right) + P' = 0 \quad (580)$$

$$\frac{(\tau+2)(\omega-1)}{2-\omega} \frac{\xi}{r} V P' + \frac{\tau}{r} P V + \left(1 + \frac{1}{r}\right) P \left(\frac{\tau+2}{2-\omega}\right) ((\omega-1)\xi V' + \omega V) = 0 \quad (581)$$

These equations can be written in the canonical form:

$$V' = \frac{\Delta_1}{\Delta}, \quad U' = \frac{\Delta_2}{\Delta}, \quad P' = \frac{\Delta_3}{\Delta} \quad (582)$$

where:

$$\Delta = a^3 \xi X - b \quad (583)$$

$$\Delta_1 = V \left(c_1 a^2 X - c_2 a - \frac{c_3}{\xi} \right) \quad (584)$$

$$\Delta_2 = -c_1 b V + c_2 a^2 X + c_3 a V \quad (585)$$

$$\Delta_3 = -c_1 abX + c_2 b + c_3 a^2 X \quad (586)$$

with

$$X = \xi V \quad (587)$$

we know that $X \rightarrow 0$ for $\xi \rightarrow 0$. Alternatively, we can solve the ODE in terms of $R = \frac{1}{V}$:

$$R' = \frac{\Delta_1}{\Delta}, \quad U' = \frac{\Delta_2}{\Delta}, \quad P' = \frac{\Delta_3}{\Delta} \quad (588)$$

$$\Delta_1 = -\left(c_1 a^2 \xi - c_2 a R - \frac{c_3}{X}\right) \quad (589)$$

$$\Delta_2 = -c_1 b \frac{1}{R} + c_2 a^2 X + c_3 a \frac{1}{R} \quad (590)$$

$$\Delta_3 = -c_1 abX + c_2 b + c_3 a^2 X \quad (591)$$

Where in both cases the coefficient are:

$$a = \frac{(\tau + 2)(1 - \omega)}{2 - \omega}, \quad b = (1 + r) a P \quad (592)$$

$$c_1 = \frac{\omega(\tau + 2)}{2 - \omega}, \quad c_2 = U \left(\frac{\omega + \tau}{2 - \omega} \right), \quad c_3 = P \left(\tau + (1 + r) \frac{(\tau + 2)\omega}{2 - \omega} \right) \quad (593)$$

Given a value of ξ_s , the ODE 582 (or 588) is integrated from $\xi = \xi_s$ to the origin $\xi = 0$. The initial conditions at the shock ξ_s are given by dimensionless Hugoniot relations:

$$V_s = \frac{r}{r + 2} ((1 - \omega) \xi_s)^{\frac{\omega}{1 - \omega}} \quad (594)$$

$$U_s = \frac{2}{r} (2 + \tau) \left(\frac{1 - \omega}{2 - \omega} \right) \xi_s V_s \quad (595)$$

$$P_s = \frac{r}{V_s} \frac{U_s^2}{2} \quad (596)$$

If the correct value of ξ_s is used, then integration of the ODE 582 from the shock to the origin, must give $P(0) = 1$. This is how ξ_s is found numerically, by finding the root of the function $P_{\xi_s}(0) - 1 = 0$ as a function of ξ_s .

The shocked mass is:

$$m_s(t) = v_0^{-\frac{1}{2 - \omega}} p_0^{\frac{1 - \omega}{2 - \omega}} t^{(2 + \tau)\frac{1 - \omega}{2 - \omega}} \xi_s \quad (597)$$

which gives a condition (we want the shock to propagate outward):

$$\tau > -2 \quad (598)$$

The piston position (which is the boundary of the system on which the external pressure is applied) as a function of time is given by:

$$x_p(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} U(0) \left(\frac{2-\omega}{2+\tau} \right) \quad (599)$$

The shock position as a function of time is given by:

$$x_s(t) = (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} ((1-\omega) \xi_s)^{\frac{1}{1-\omega}} \quad (600)$$

The position of an element with Lagrangian coordinate m , as a function of time is given by:

$$x(t, m) = \begin{cases} (v_0 p_0 t^{2+\tau})^{\frac{1}{2-\omega}} \left[(1-\omega) \xi V(\xi) + \left(\frac{2-\omega}{\tau+2} \right) U(\xi) \right] & m < m_s(t) \\ [v_0 (1-\omega) m]^{\frac{1}{1-\omega}} & \text{else} \end{cases} \quad (601)$$

where $\xi(m)$ is given by eq. 575. This relation is obtained by a nice identity for the V integral:

$$\int_0^\xi V(\xi') d\xi' = q_1 \xi V(\xi) + q_2 (U(\xi) - U(0)), \quad (602)$$

where:

$$q_1 = 1 - \omega, \quad q_2 = \frac{2 - \omega}{\tau + 2} \quad (603)$$

We note that for $m = m_s(t)$, both expressions in eq. 601 (shocked and unshocked regions) coincides exactly (by using the Jump conditions in eq. 602). Internal energy:

$$E_{in}(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \frac{P(\xi) V(\xi)}{r} d\xi \quad (604)$$

kinetic energy:

$$E_k(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \frac{U^2(\xi)}{2} d\xi \quad (605)$$

we have the identity:

$$\frac{E_k(t)}{E_{in}(t)} = \text{const} \quad (606)$$

total energy:

$$E(t) = E_{in}(t) + E_k(t) = v_0^{\frac{1}{2-\omega}} p_0^{\frac{3-\omega}{2-\omega}} t^{\frac{3\tau-\tau\omega+2}{2-\omega}} \int_0^{\xi_s} \left(\frac{P(\xi) V(\xi)}{r} + \frac{U^2(\xi)}{2} \right) d\xi \quad (607)$$

we have the exact identity for any ξ :

$$\int_0^\xi \left(\frac{P(\xi') V(\xi')}{r} + \frac{1}{2} U^2(\xi') \right) d\xi' = \frac{(2+\tau)(1-\omega)}{3\tau - \tau\omega + 2} \xi \left(\frac{P(\xi) V(\xi)}{r} + \frac{1}{2} U^2(\xi) \right) \quad (608)$$

$$- \frac{2-\omega}{3\tau - \tau\omega + 2} (P(\xi) U(\xi) - U(0)) \quad (609)$$

For the total integral on $\xi = \xi_s$ this is reduced to:

$$\int_0^{\xi_s} \left(\frac{P(\xi) V(\xi)}{r} + \frac{1}{2} U^2(\xi) \right) d\xi = \frac{2-\omega}{3\tau - \tau\omega + 2} U(0) \quad (610)$$